

# Exercises — MDP, Return, Value Functions, Bellman Equations

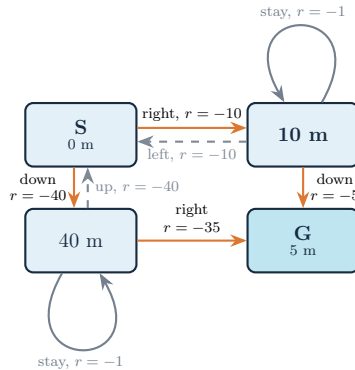
Lecture 1/4 — Reinforcement Learning

CentraleSupélec

## Topics covered

- **Formalizing an MDP:**  $\mathcal{S}$ ,  $\mathcal{A}(s)$ ,  $p(s', r \mid s, a)$ .
- **Policies**, deterministic and stochastic, normalization condition.
- **Return**  $G_t$  and the role of the discount factor  $\gamma$ .
- **Value functions**  $v_\pi$ ,  $q_\pi$ , and their relation.
- **Bellman equations**, expectation and optimality.
- **Proofs** (Exercises 9–10): deriving the Bellman equations from the definition of  $G_t$ , and the effect of shifting all rewards by a constant.
- **Going further** (Exercises 11–15, from Sutton & Barto): the role of  $\gamma$  in choosing  $\pi_*$ , the  $v_*/q_*/\pi_*$  dictionary, three-argument notation, episodic versus continuing tasks, and the limits of the MDP framework.

**Reference environment (recall from lecture).** Four states:  $S$  (start, 0 m),  $A$  (10 m),  $B$  (40 m),  $G$  (terminal, 5 m). Reward  $r(s, a) = -|\text{altitude}(s') - \text{altitude}(s)|$ : this is a **cost-of-effort** MDP (minimize total climbing effort to reach  $G$ ), not a sparse goal-bonus one — there is no reward for arriving at  $G$  beyond the cost of the last step.  $\gamma = 1$  unless stated otherwise. *Note:* the *stay* self-loop costs  $r = -1$  (not free) — with  $\gamma = 1$ , a free ( $r = 0$ ) self-loop would make an infinite stay tie with any finite-cost path to  $G$ , an unbounded-return degeneracy. At  $r = -1$ , lingering forever costs  $-\infty$ , so *stay* is always dominated and never appears in an optimal policy (see Exercise 7) — no special-casing needed.



### Exercise 1 — Formalize the environment as an MDP

- (a) List  $\mathcal{S}$  and  $\mathcal{A}(s)$  for each of the four states (careful:  $G$  is terminal,  $\mathcal{A}(G) = \emptyset$ ).
- (b) This environment is **deterministic**: for a given pair  $(s, a)$ , exactly one pair  $(s', r)$  has nonzero probability. Write  $p(s', r \mid s, a)$  for the three pairs  $(S, \text{right})$ ,  $(A, \text{stay})$  and  $(B, \text{up})$ .
- (c) Check that  $\sum_{s', r} p(s', r \mid S, \text{right}) = 1$ . Why does this sum always equal 1, regardless of the environment?

### Exercise 2 — Two policies

Consider two policies on this environment:

- $\pi_A$  (**deterministic**): always *right* at  $S$ , always *down* at  $A$ , always *right* at  $B$ .
  - $\pi_B$  (**stochastic**): at  $S$ ,  $\pi(\text{right} \mid S) = 0.3$  and  $\pi(\text{down} \mid S) = 0.7$ ; elsewhere identical to  $\pi_A$  (deterministic at  $A$  and  $B$ ).
- (a) Write  $\pi(a \mid S)$  for  $\pi_A$  and for  $\pi_B$  as a table (one row per possible action at  $S$ ).
  - (b) Check  $\sum_a \pi(a \mid S) = 1$  for both policies.
  - (c) Which of the two matches the “Dirac mass” mentioned in lecture (slide “Policy”)? Write  $\pi_A(\cdot \mid S)$  in that form.

### Exercise 3 — Return and discount factor

Under  $\pi_A$ , the unique trajectory from  $S$  is

$$S \xrightarrow[R_1=-10]{\text{right}} A \xrightarrow[R_2=-5]{\text{down}} G \text{ (end)}.$$

- (a) Compute the return  $G_0$  (with  $\gamma = 1$ ).
- (b) Recompute  $G_0$  with  $\gamma = 0.9$ .
- (c) Compute  $G_1$  (the return from state  $A$  onward) in both cases, and check the relation  $G_0 = R_1 + \gamma G_1$ .
- (d) In one sentence: how does  $\gamma$  change the relative weight of  $R_1$  (immediate) versus  $R_2$  (future) in  $G_0$ ?

### Exercise 4 — Value functions under $\pi_A$

Stay under  $\pi_A$ , with  $\gamma = 1$ .

- (a) By backward substitution from  $G$  (where  $v_\pi(G) = 0$ ), compute  $v_{\pi_A}(A)$  then  $v_{\pi_A}(S)$ .
- (b) Compute  $v_{\pi_A}(B)$  (recall:  $\pi_A$  also goes *right* from  $B$ ).
- (c) Compute  $q_{\pi_A}(S, \text{right})$  and  $q_{\pi_A}(S, \text{down})$ .
- (d) Compute  $q_{\pi_A}(A, \text{down})$  and  $q_{\pi_A}(A, \text{left})$  (the latter leads back to  $S$ : use the value  $v_{\pi_A}(S)$  found in (a)).

- (e) *Check*: compare  $q_{\pi_A}(S, \text{right})$  and  $q_{\pi_A}(S, \text{down})$  to the values  $q_{\pi}(S, \text{right}) = -15$  and  $q_{\pi}(S, \text{down}) = -75$  from the “Action-value function - example” slide in lecture. Is this consistent? Why (the lecture’s policy isn’t  $\pi_A$ )?

**Exercise 5 —  $v_{\pi}$  as an average of  $q_{\pi}$  under  $\pi_B$**

- (a) Using the values of  $q_{\pi_A}(S, \text{right})$  and  $q_{\pi_A}(S, \text{down})$  found in the previous exercise (still valid under  $\pi_B$ : why?), and the probabilities  $\pi(\cdot | S)$  of  $\pi_B$  (Exercise 2), compute  $v_{\pi_B}(S)$  using

$$v_{\pi}(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) q_{\pi}(s, a).$$

- (b) Is the result closer to  $q_{\pi_A}(S, \text{right})$  or to  $q_{\pi_A}(S, \text{down})$ ? Is this consistent with the 0.3 / 0.7 probabilities chosen?

**Exercise 6 — Bellman expectation equation**

- (a) Write the general Bellman expectation equation for  $v_{\pi}(S)$  (sum over actions, then over  $(s', r)$ ).
- (b) Instantiate it for  $\pi_B$  at  $S$  (two possible actions, deterministic environment so the sum over  $(s', r)$  has a single nonzero term per action).
- (c) Check that it reproduces  $v_{\pi_B}(S)$  found in the previous exercise.

**Exercise 7 — Optimal value function and optimal policy**

Now  $A$  and  $B$  each have 3 actions available: the two moves already used, plus *stay* ( $r = -1$ ).

- (a) For  $A$  and for  $B$ , compare all 3 available actions and justify that both *stay* and going back toward  $S$  are never optimal. Deduce  $v_*(A)$  and  $v_*(B)$ .
- (b) Deduce  $v_*(S)$  (compare *right* and *down* from  $S$ , using  $v_*(A)$  and  $v_*(B)$ ).
- (c) *Consistency check*: plug  $v_*(S)$  back into the equations of (a) and check they still hold (fixed point of the optimality equation).
- (d) Deduce  $\pi_*$ . Compare to  $\pi_A$  (Exercises 4 and 5) and to  $v_{\pi_A}(S)$  found in Exercise 4(a). What does the existence theorem for a deterministic optimal policy say about this result?

**Exercise 8 — Returns from a reward sequence (classic, Sutton & Barto Ch. 3)**

- (a) **Episodic task**. Suppose  $\gamma = 0.5$  and the following sequence of rewards is received:  $R_1 = -1$ ,  $R_2 = 2$ ,  $R_3 = 6$ ,  $R_4 = 3$ ,  $R_5 = 2$ , with the episode ending at  $T = 5$  (so  $G_5 = 0$ ). Compute  $G_0, G_1, \dots, G_5$ . Work **backwards**, using  $G_t = R_{t+1} + \gamma G_{t+1}$ .
- (b) **Continuing task**. Now suppose  $\gamma = 0.9$  and the reward sequence is  $R_1 = 2$  followed by an **infinite** sequence of 7s (i.e.  $R_2 = R_3 = \dots = 7$ ). Compute  $G_1$  using the geometric series identity  $\sum_{k=0}^{\infty} \gamma^k = \frac{1}{1-\gamma}$  (valid for  $\gamma < 1$ ), then  $G_0$ .
- (c) Episode (a) terminates after 5 steps; the reward sequence in (b) never ends. Why is  $G_1$  still a **finite** number in (b)?

### Exercise 9 — Deriving the Bellman equations (theory bonus, Sutton & Barto Ch. 3)

Exercise 6 *used* the Bellman expectation equation; this one **derives** it, from the definition of the return and nothing else. Throughout,  $\pi$  is an arbitrary policy and all expectations are taken over trajectories generated by  $\pi$ .

- (a) **From  $v_\pi$  down to  $q_\pi$ .** Starting from  $v_\pi(s) \doteq \mathbb{E}_\pi[G_t | S_t = s]$  and conditioning on the action taken at time  $t$ , prove

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) q_\pi(s, a).$$

*Hint:* law of total expectation, splitting on the value of  $A_t$ ; the probability of  $A_t = a$  given  $S_t = s$  is precisely  $\pi(a | s)$ .

- (b) **From  $q_\pi$  down to  $v_\pi$ .** Using the recursion  $G_t = R_{t+1} + \gamma G_{t+1}$  and conditioning this time on the pair  $(S_{t+1}, R_{t+1})$ , prove

$$q_\pi(s, a) = \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')].$$

At which exact step do you need the **Markov property**? State precisely what it is used for.

- (c) **Assembling.** Substitute (b) into (a) to recover the Bellman expectation equation for  $v_\pi$  — the one Exercise 6 took for granted. Then substitute (a), written at  $s'$ , into (b) to obtain the Bellman expectation equation for  $q_\pi$ , i.e.  $q_\pi(s, a)$  in terms of the  $q_\pi(s', a')$  of its successors.
- (d) **Optimality.** Admit that an optimal policy exists. Show that  $v_*(s) = \max_{a \in \mathcal{A}(s)} q_*(s, a)$ , and explain why the  $\sum_a \pi(a | s)$  of (a) becomes a  $\max_a$  here, while the sum over  $(s', r)$  in (b) does **not**. Deduce the Bellman optimality equation for  $v_*$ .

### Exercise 10 — Do the signs of the rewards matter? (theory bonus, Sutton & Barto Ch. 3)

Every reward in this sheet's environment is **negative**, which may feel like an arbitrary modelling choice: why not add a constant to make them all positive, and maximize a “score” instead of minimizing a cost? This exercise settles the question — and the answer is not the same for continuing and for episodic tasks.

Fix a constant  $c \in \mathbb{R}$  and define a new MDP, identical to the old one except that every reward is shifted:  $R'_t = R_t + c$ . Write  $v'_\pi$  for value functions in the shifted MDP.

- (a) **Continuing task,  $\gamma < 1$ .** Prove that there is a constant  $v_c$ , *independent of the state and of the policy*, such that

$$v'_\pi(s) = v_\pi(s) + v_c \quad \text{for every } s \text{ and every } \pi,$$

and give  $v_c$  in terms of  $c$  and  $\gamma$ . *Hint:* start from  $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$  and use the geometric series of Exercise 8(b).

- (b) Deduce that the shift changes neither the ordering of policies nor  $\pi_*$ . Conclude: in a continuing task, only the **differences** between rewards carry information, not their signs.

- (c) **Episodic task**,  $\gamma = 1$ . Let  $T$  be the (random) time at which the episode ends. Show that now

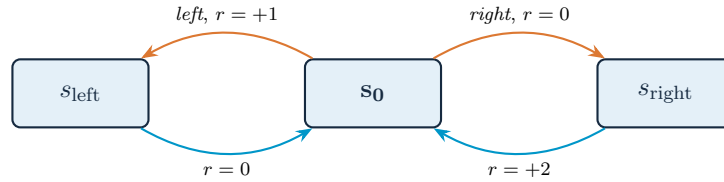
$$v'_\pi(s) = v_\pi(s) + c \mathbb{E}_\pi[T \mid S_0 = s],$$

and explain in one sentence why this is **not** the situation of (a).

- (d) **Counterexample on our own environment**. Take the environment of this sheet ( $\gamma = 1$ , *stay* available at  $A$  and  $B$  with  $r = -1$ ) and shift every reward by  $c = +11$ . Compare the policy  $\pi_A$  ( $S \rightarrow A \rightarrow G$ , reaching  $G$  in 2 steps) with the policy that goes to  $A$ , plays *stay*  $k$  times, then goes to  $G$ . Compute both returns from  $S$  in the shifted MDP as functions of  $k$ , and conclude. What is  $v'_*(S)$ ?
- (e) For which values of  $c$  does the shifted task stay well-posed? Relate your answer to the note on the *stay* self-loop at the top of this sheet ( $r = -1$  rather than  $r = 0$ ), and to Exercise 7(a).

### Exercise 11 — The discount factor decides the optimal policy (Sutton & Barto 3.22)

Every environment so far has been **episodic**. Here is a **continuing** one — it never terminates. The only decision is taken in the top state  $s_0$ , where two actions are available; every other transition is forced. Rewards are deterministic, shown on the arrows.



In  $s_{\text{left}}$  and  $s_{\text{right}}$  a single action is available (the return arrow below), so the agent decides only in  $s_0$ : there are exactly two deterministic policies,  $\pi_{\text{left}}$  and  $\pi_{\text{right}}$ .

- (a) Write the infinite reward sequence seen from  $s_0$  under each of the two policies.
- (b) Show that

$$v_{\pi_{\text{left}}}(s_0) = \frac{1}{1 - \gamma^2}, \quad v_{\pi_{\text{right}}}(s_0) = \frac{2\gamma}{1 - \gamma^2}.$$

*Hint:* each policy returns to  $s_0$  every 2 steps, so group the rewards in pairs — the series is geometric with ratio  $\gamma^2$ .

- (c) Which policy is optimal for  $\gamma = 0$ ? For  $\gamma = 0.9$ ? For  $\gamma = 0.5$ ?
- (d) Solve  $v_{\pi_{\text{left}}}(s_0) = v_{\pi_{\text{right}}}(s_0)$  exactly. What does this exercise say about  $\gamma$  that Exercise 3(d) did not?

### Exercise 12 — The $v_*$ / $q_*$ / $\pi_*$ dictionary (Sutton & Barto 3.25–3.28)

Four one-line equations, used constantly from Lecture 2 on. Give each, and justify it in a sentence.

- (a)  $v_*$  in terms of  $q_*$ .
- (b)  $q_*$  in terms of  $v_*$  and the four-argument  $p(s', r \mid s, a)$ .

- (c)  $\pi_*$  in terms of  $q_*$ .
- (d)  $\pi_*$  in terms of  $v_*$  and the four-argument  $p$ .
- (e) **The point.** Compare (c) and (d): to act optimally from  $q_*$ , what must you know about the environment? And from  $v_*$ ? Deduce why the algorithms in the second half of this course estimate  $q$  rather than  $v$ .

**Exercise 13 — The same equations in three-argument notation (Sutton & Barto 3.29)**

Much of the literature uses not the four-argument  $p(s', r | s, a)$ , but the pair

$$p(s' | s, a) \doteq \sum_{r \in \mathcal{R}} p(s', r | s, a), \quad r(s, a) \doteq \mathbb{E}[R_{t+1} | S_t = s, A_t = a] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}^+} p(s', r | s, a).$$

- (a) Show that for any function  $v$ ,

$$\sum_{s', r} p(s', r | s, a) [r + \gamma v(s')] = r(s, a) + \gamma \sum_{s'} p(s' | s, a) v(s').$$

Which property of the bracket makes this work?

- (b) Deduce the four Bellman equations — for  $v_\pi$ ,  $q_\pi$ ,  $v_*$  and  $q_*$  — written with  $p(s' | s, a)$  and  $r(s, a)$  only.
- (c) In one sentence: what is lost, and what is gained, in this notation?

**Exercise 14 — Episodic or continuing? (Sutton & Barto 3.5 and 3.6)**

- (a) The normalization condition of Exercise 1(c),

$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) = 1 \quad \text{for all } s \in \mathcal{S}, a \in \mathcal{A}(s),$$

is written for the **continuing** case. Give the (very slight) modification needed for an **episodic** task, and say why it is needed. *Hint:* where does the agent land on the last step of an episode?

- (b) Pole-balancing, treated as an **episodic** task but *with* discounting: all rewards are 0 except  $-1$  upon failure. Let  $K$  be the number of steps from  $t$  until the next failure. Give  $G_t$ .
- (c) The same task in the **continuing** formulation (the pole is reset after each failure and the task never ends). Give  $G_t$  now, and say how it differs from (b).
- (d) Check in both cases that maximizing the return really does mean “balance the pole as long as possible”.

**Exercise 15 — Limits of the MDP framework (discussion, Sutton & Barto 3.2 and 3.3)**

No computation — two questions worth a genuine discussion, with no unique answer.

- (a) Is the MDP framework sufficient to usefully represent *all* goal-directed learning tasks? Propose at least two clear counterexamples, saying in each case what exactly breaks.
- (b) Consider driving a car. The actions could be defined at the level of the accelerator, steering wheel and brakes; or one level below (the driver’s muscles); or one level above (the choice of route). Where should the boundary between **agent** and **environment** be drawn? Is there a single right answer? What rule of thumb would you propose?

## Notation & formulae

- **MDP:**  $\mathcal{S}$  (states),  $\mathcal{A}(s)$  (actions available at  $s$ ),  $p(s', r \mid s, a) \doteq \mathbb{P}[S_t = s', R = r \mid S_{t-1} = s, A_{t-1} = a]$ .
- **Policy:**  $\pi(a \mid s) \doteq \mathbb{P}[A_t = a \mid S_t = s]$ , with  $\sum_{a \in \mathcal{A}(s)} \pi(a \mid s) = 1$  (or  $\int_{\mathcal{A}(s)} \pi(a \mid s) da = 1$  for continuous actions). Deterministic  $\pi$ :  $\pi(\cdot \mid s) = \delta_{\pi(s)}$ .
- **Return:**  $G_t \doteq \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = R_{t+1} + \gamma G_{t+1}$ .
- **Value functions:**  $v_{\pi}(s) \doteq \mathbb{E}_{\pi}[G_t \mid S_t = s]$ ,  $q_{\pi}(s, a) \doteq \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a]$ .
- $v_{\pi}$  as an average of  $q_{\pi}$ :  $v_{\pi}(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) q_{\pi}(s, a)$ .
- **Bellman expectation equations:**

$$v_{\pi}(s) = \sum_a \pi(a \mid s) \sum_{s', r} p(s', r \mid s, a) [r + \gamma v_{\pi}(s')], \quad q_{\pi}(s, a) = \sum_{s', r} p(s', r \mid s, a) \left[ r + \gamma \sum_{a'} \pi(a' \mid s') q_{\pi}(s', a') \right]$$

- **Optimal value functions:**  $v_*(s) \doteq \max_{\pi} v_{\pi}(s)$ ,  $q_*(s, a) \doteq \max_{\pi} q_{\pi}(s, a)$ .
- **Bellman optimality equations:**

$$v_*(s) = \max_{a \in \mathcal{A}(s)} \sum_{s', r} p(s', r \mid s, a) [r + \gamma v_*(s')], \quad q_*(s, a) = \sum_{s', r} p(s', r \mid s, a) \left[ r + \gamma \max_{a'} q_*(s', a') \right]$$

- **Optimal policy:**  $\pi_*(s) \doteq \arg \max_{a \in \mathcal{A}(s)} q_*(s, a)$ . At least one deterministic  $\pi_*$  always exists (finite MDP, bounded rewards).